

We consider a smooth three-dimensional function

$$f(x) = \frac{1}{1 + \|Cx\|^2},$$

where  $x \in \mathbb{R}^3$  and  $C \in \mathbb{R}^{3 \times 3}$  controls anisotropy.

Writing  $x = (x_1, x_2, x_3)$ , the function admits a Chebyshev expansion

$$f(x_1, x_2, x_3) = \sum_{k=0}^{d_x} \sum_{\ell=0}^{d_y} \sum_{m=0}^{d_z} C_{k\ell m}^{\text{true}} T_k(x_1) T_\ell(x_2) T_m(x_3),$$

where  $T_k$  denotes the Chebyshev polynomial of degree  $k$  and  $C^{\text{true}}$  is the reference coefficient tensor obtained by projecting  $f$  onto the tensor-product Chebyshev basis on a tensor grid.

For each configuration  $s = 1, \dots, S$ , we generate a set of points

$$X^{(s)} = \{x_1^{(s)}, \dots, x_n^{(s)}\}, \quad x_i^{(s)} \in [-1, 1]^3.$$

Define the scalar energy as the mean over the points in the configuration

$$E^{(s)} = \frac{1}{n} \sum_{i=1}^n f(x_i^{(s)}).$$

Writing

$$x_i^{(s)} = (x_{i1}^{(s)}, x_{i2}^{(s)}, x_{i3}^{(s)}),$$

we define pooled ACE features

$$a_1^{(s)}(k) = \sum_{i=1}^n T_k(x_{i1}^{(s)}), \quad a_2^{(s)}(\ell) = \sum_{i=1}^n T_\ell(x_{i2}^{(s)}), \quad a_3^{(s)}(m) = \sum_{i=1}^n T_m(x_{i3}^{(s)}).$$

By linearity of the expansion, the energy can be written as

$$E^{(s)} = \sum_{k, \ell, m} \frac{1}{n} C_{k\ell m}^{\text{true}} a_1^{(s)}(k) a_2^{(s)}(\ell) a_3^{(s)}(m).$$

We therefore fit a trilinear model

$$E^{(s)} \approx \sum_{k, \ell, m} C_{k\ell m} a_1^{(s)}(k) a_2^{(s)}(\ell) a_3^{(s)}(m),$$

Approach 1:

$$C \approx G \times_1 A \times_2 B \times_3 D,$$

with

$$A \in \mathbb{R}^{(d_x+1) \times R_x}, \quad B \in \mathbb{R}^{(d_y+1) \times R_y}, \quad D \in \mathbb{R}^{(d_z+1) \times R_z},$$

and core tensor

$$G \in \mathbb{R}^{R_x \times R_y \times R_z}.$$

The parameters  $A$ ,  $B$ ,  $D$ , and  $G$  are computed by alternating least squares (ALS).

The factor matrices are averaged

$$U = \frac{1}{3}(A + B + D),$$

and then set

$$A = B = D = U.$$

The core tensor is also symmetrized

$$G_{pqr} = \frac{1}{6} \left( G_{pqr} + G_{prq} + G_{qpr} + G_{qrp} + G_{rpq} + G_{rqp} \right)$$

### Approach 2:

Instead of first computing independent factor matrices and then averaging them, we directly impose symmetry in the Tucker model by using a single shared factor matrix

$$A \in \mathbb{R}^{(d+1) \times R}.$$

The coefficient tensor is approximated as

$$C \approx G \times_1 A \times_2 A \times_3 A,$$

where

$$G \in \mathbb{R}^{R \times R \times R}$$

is a supersymmetric core tensor.

For each configuration  $s$ , we define the reduced features

$$x^{(s)} = A_1^{(s)} A, \quad y^{(s)} = A_2^{(s)} A, \quad z^{(s)} = A_3^{(s)} A,$$

where  $A_1^{(s)}, A_2^{(s)}, A_3^{(s)}$  denote the pooled ACE feature vectors.

The predicted energy is then

$$\hat{E}^{(s)} = \sum_{p,q,r} G_{pqr} x_p^{(s)} y_q^{(s)} z_r^{(s)}.$$

To update the shared factor matrix, we construct a least-squares tensor

$$\tilde{G}_{kqr},$$

$$\tilde{G}_{kqr} = \arg \min_{\tilde{G}} \sum_{s=1}^S \left( E^{(s)} - \sum_{k,q,r} \tilde{G}_{kqr} a_1^{(s)}(k) y_q^{(s)} z_r^{(s)} \right)^2$$

and reshape it into a matrix

$$\tilde{G}_{(1)} \in \mathbb{R}^{(d+1) \times (R^2)}.$$

We compute its singular value decomposition

$$\tilde{G}_{(1)} = A_{\text{svd}} \Sigma V^\top.$$

The updated factor matrix is obtained from the leading  $R$  singular vectors

$$A = A_{\text{svd}}(:, 1:R).$$

The core tensor is obtained from the same SVD through the unfolded core

$$G_{(1)} = \Sigma_R V_R^\top,$$

which is reshaped into a tensor

$$G \in \mathbb{R}^{R \times R \times R}.$$

Finally the core tensor is symmetrized

$$G_{pqr} = \frac{1}{6} (G_{pqr} + G_{prq} + G_{qpr} + G_{qrp} + G_{rqp} + G_{rpq}).$$

### Standard Symmetric Tucker Decomposition

For a symmetric third-order tensor

$$C \in \mathbb{R}^{d \times d \times d},$$

the symmetric Tucker decomposition uses a single factor matrix

$$A \in \mathbb{R}^{d \times R}.$$

The tensor is represented as

$$C \approx G \times_1 A \times_2 A \times_3 A,$$

where

$$G \in \mathbb{R}^{R \times R \times R}$$

is a symmetric core tensor.

The decomposition is computed by alternating least squares, updating  $A$  and then projecting to obtain the core

$$G = C \times_1 A^\top \times_2 A^\top \times_3 A^\top.$$

